Elementary Analysis

Chapter 6, Sections 32–36

All 62 exercises, with every subpart
Detailed worked solutions

Mathjock

Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Sections 32–36, 62 exercises. Short descriptions identify the problems; consult the linked second-edition text for the original statements.

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32 The Riemann integral

For a bounded function f and a partition $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$, let $I_j = [t_{j-1}, t_j]$. The lower and upper sums are

$$L(f,P) = \sum_j (\inf_{I_j} f)\,|I_j|\,, \qquad U(f,P) = \sum_j (\sup_{I_j} f)\,|I_j|\,,$$

where $|I_j| = t_j - t_{j-1}$ denotes interval length. The lower integral is the supremum of all lower sums; the upper integral is the infimum of all upper sums. Write these as $\underline{I}(f)$ and $\overline{I}(f)$. Integrability means they are equal. A practical equivalent test is that $U(f,P) - L(f,P)$ can be made arbitrarily small. No antiderivative formula is assumed in the Darboux calculations below.

32.1 Exercise 32.1: The integral of a cube from sums

What we must show. Compute both Darboux integrals of x^3 on $[0,b]$, where $b > 0$.

Choose equal subintervals and identify maxima and minima.

The cube is increasing on $[0,b]$. For $t_k = kb/n$, its maximum on $[t_{k-1}, t_k]$ is t_k^3 and its minimum is t_{k-1}^3. Therefore

$$U_n = \frac{b^4}{n^4}\sum_{k=1}^{n} k^3 = \frac{b^4(n+1)^2}{4n^2}, \qquad L_n = \frac{b^4}{n^4}\sum_{k=0}^{n-1} k^3 = \frac{b^4(n-1)^2}{4n^2}.$$

Here we used $\sum_{k=1}^{n} k^3 = [n(n+1)/2]^2$.

Squeeze the integrals.

For every n, $L_n \le \underline{I}(f) \le \overline{I}(f) \le U_n$. Both outer sequences tend to $b^4/4$, because $(n \pm 1)^2/n^2 \to 1$. Thus the two integrals coincide at that value.

Conclusion. The function is integrable and $\int_0^b x^3\,dx = b^4/4$.

32.2 Exercise 32.2: A rational-only linear function

What we must show. For $f(x) = x$ on rational points and $f(x) = 0$ on irrational points of $[0,b]$, compute its lower and upper integrals.

Use density on each interval.

Every nondegenerate interval $[u,v] \subseteq [0,b]$ contains irrational points, so its infimum is zero. Its rational points can approach v from below; their values approach v, while no value exceeds v. Thus its supremum is v, whether or not v itself is rational. For any partition, consequently,

$$L(f,P) = 0, \qquad U(f,P) = \sum_{j=1}^{n} t_j(t_j - t_{j-1}).$$

In particular, the lower integral is 0.

Find the infimum of the upper sums.

Since $t_j \geq t_{j-1} \geq 0$,

$$t_j(t_j - t_{j-1}) \geq \frac{t_j + t_{j-1}}{2}(t_j - t_{j-1}) = \frac{t_j^2 - t_{j-1}^2}{2}.$$

Summing telescopes to $U(f, P) \geq b^2/2$. For the equal partition,

$$U(f, P_n) = \frac{b^2}{n^2} \sum_{j=1}^{n} j = \frac{b^2(n+1)}{2n} \longrightarrow \frac{b^2}{2}.$$

An infimum bounded below by $b^2/2$ and approached by such sums equals $b^2/2$.

Conclusion. (a) The lower integral is 0 and the upper integral is $b^2/2$. (b) They differ for $b > 0$, so f is not integrable.

32.3 Exercise 32.3: A rational-only quadratic function

What we must show. Repeat the calculation for $g(x) = x^2$ on rational points and $g(x) = 0$ on irrational points of $[0, b]$.

Identify each interval's extrema.

Density and nonnegativity give $\inf_{[u,v]} g = 0$ and $\sup_{[u,v]} g = v^2$. Thus all lower sums vanish and $U(g, P) = \sum_j t_j^2(t_j - t_{j-1})$.

Compute the infimum of these upper sums.

For $0 \leq u \leq v$ we have $v^2 \geq (v^2 + vu + u^2)/3$, whence

$$v^2(v - u) \geq \frac{v^3 - u^3}{3}.$$

This telescopes to $U(g, P) \geq b^3/3$ for every P. For equal subintervals,

$$U(g, P_n) = \frac{b^3}{n^3} \sum_{j=1}^{n} j^2 = \frac{b^3 n(n+1)(2n+1)}{6n^3} \longrightarrow \frac{b^3}{3}.$$

Conclusion. (a) The lower and upper integrals are 0 and $b^3/3$. (b) The function is not integrable on $[0, b]$ for $b > 0$.

32.4 Exercise 32.4: Refinement by induction

What we must show. Extend the one-point refinement argument in Lemma 32.2 to any finite refinement $P \subseteq Q$.

State precisely what is being inducted on.

Let r be the number of points of Q not already in P. We claim

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P).$$

For $r = 0$ the partitions are equal; the middle inequality follows because each interval's infimum is at most its supremum.

Insert one more point.

Suppose the result is established for r extra points. If Q has $r+1$ extra points, choose one of them, q, and set $Q' = Q \setminus \{q\}$. By the induction hypothesis, $L(f,P) \le L(f,Q')$ and $U(f,Q') \le U(f,P)$. The proved one-point case, applied to Q' and Q, gives $L(f,Q') \le L(f,Q)$ and $U(f,Q) \le U(f,Q')$. Combining the inequalities proves the two outer comparisons for Q. The middle comparison always holds interval by interval.

Conclusion. Adding finitely many partition points can only increase a lower sum and decrease an upper sum.

32.5 Exercise 32.5: Why the lower integral cannot exceed the upper

What we must show. Apply the order property of suprema and infima to Theorem 32.4.

Name the two sets.

Let

$$S = \{L(f,P) : P \text{ is a partition of } [a,b]\}, \qquad T = \{U(f,Q) : Q \text{ is a partition of } [a,b]\}.$$

Both are nonempty and bounded because f is bounded and the total interval length is $b-a$.

Compare sums from different partitions.

For any P, Q, their union $R = P \cup Q$ refines both. Lemma 32.2 gives

$$L(f,P) \le L(f,R) \le U(f,R) \le U(f,Q).$$

Thus every $s \in S$ is at most every $t \in T$. By Exercise 4.8, $\sup S \le \inf T$. To see the order reasoning directly, each $t \in T$ is an upper bound for S, so $\sup S \le t$ for all $t \in T$; therefore $\sup S$ is a lower bound for T, and hence is at most $\inf T$.

Conclusion. $\underline{I}(f) = \sup S \le \inf T = \overline{I}(f)$.

32.6 Exercise 32.6: Upper and lower sums whose gap tends to zero

What we must show. Prove integrability and identify the limits of the given sums. The sums U_n and L_n need not come from the same partition. Nevertheless, the cross-partition comparison just established gives

$$L_n \le \underline{I}(f) \le \overline{I}(f) \le U_n.$$

Subtracting shows $0 \le \overline{I}(f) - \underline{I}(f) \le U_n - L_n \to 0$. The fixed middle difference must therefore be zero. Write the common integral as I. Then

$$0 \le U_n - I \le U_n - L_n, \qquad 0 \le I - L_n \le U_n - L_n.$$

Each right-hand side tends to zero, so both $U_n \to I$ and $L_n \to I$.

Conclusion. f is integrable and $\int_a^b f = \lim_n U_n = \lim_n L_n$.

32.7 Exercise 32.7: Changing finitely many values

What we must show. Show that changing an integrable function at finitely many points preserves its integral.

First understand the finite-support error.

Put $h = g - f$. It is zero except at finitely many points $c_1, \dots, c_m$, so it is bounded; choose $H > 0$ with $|h| \le H$. Given any $\eta > 0$, make a partition that contains the exceptional points and nearby points on each available side, so the total length of cells meeting these points is less than η. This is possible because there are only finitely many points. If a point is a or b, only its one adjacent side is needed. On every other cell $h = 0$; on the exceptional cells $-H \le h \le H$. Thus

$$-H\eta \le L(h, P) \le U(h, P) \le H\eta.$$

This proves that a finite-support function has lower and upper integrals both zero, by letting $\eta \downarrow 0$.

Combine the estimate with the original integrable function.

Here is a direct argument that does not assume a later linearity theorem. Given $\varepsilon > 0$, choose a partition for f with upper-minus-lower gap less than $\varepsilon/2$. Refine it to the partition just described, with exceptional-cell total length less than η and $2H\eta < \varepsilon/2$. Refinement preserves the bound for f. On any cell J, $g = f + h$ implies

$$\sup_J g \le \sup_J f + \sup_J h, \qquad \inf_J g \ge \inf_J f + \inf_J h.$$

Consequently $U(g, P) - L(g, P) < \varepsilon/2 + 2H\eta < \varepsilon$. This proves g is integrable.

Compare the two integral values.

For the same partition, $L(g, P) \ge L(f, P) - H\eta$ and $U(g, P) \le U(f, P) + H\eta$. Both integrals lie between the respective sums. Hence their difference in absolute value is at most $U(f, P) - L(f, P) + H\eta$. The original gap and η can both be made arbitrarily small.

Conclusion. $\int_a^b g = \int_a^b f$. Isolated values do not change area.

32.8 Exercise 32.8: Restriction to a smaller interval

What we must show. Prove integrability of f on $[c, d] \subseteq [a, b]$. Assume $c < d$; the zero-length case has integral zero by convention. Given $\varepsilon > 0$, choose a partition P of $[a, b]$ with $U(f, P) - L(f, P) < \varepsilon$. Add c and d to form the refinement Q. Its gap is still less than ε. The points of Q lying in $[c, d]$ form a partition Q_0 of that interval. Its gap is a sub-sum of the gap for Q:

$$U_{[c,d]}(f, Q_0) - L_{[c,d]}(f, Q_0) = \sum_{J \subseteq [c,d]} (\sup_J f - \inf_J f)\,|J| \le U_{[a,b]}(f, Q) - L_{[a,b]}(f, Q).$$

Every term omitted on the right is nonnegative. Thus the smaller interval has partitions with gap below every positive ε.

Conclusion. The Darboux criterion proves that $f|_{[c,d]}$ is integrable.

33 Properties of the Riemann integral

The Darboux criterion converts integrability into an estimate of the total oscillation of a function over short intervals. Once integrability has been established, linearity, comparison, and interval additivity make calculations much easier. Keep two issues separate: first justify that an integral exists; then use its algebraic properties. A pointwise limit alone does not usually allow us to move a limit through an integral.

33.1 Exercise 33.1: Decreasing functions are integrable

What we must show. Complete the monotone-function integrability proof. Let f be decreasing on $[a, b]$, meaning nonincreasing. It is bounded between $f(b)$ and $f(a)$. If these values agree, f is constant and every upper and lower sum agrees. Otherwise $D = f(a) - f(b) > 0$. For a partition $P = \{t_0, \dots, t_n\}$, decreasingness gives

$$U(f, P) - L(f, P) = \sum_{j=1}^{n} [f(t_{j-1}) - f(t_j)](t_j - t_{j-1}) \le \operatorname{mesh}(P) \sum_{j=1}^{n} [f(t_{j-1}) - f(t_j)].$$

The last sum telescopes to D. Given $\varepsilon > 0$, choose an equal partition with mesh less than ε/D. Then the gap is below ε.

Conclusion. The Darboux criterion proves integrability, including the constant case.

33.2 Exercise 33.2: Scaling suprema and infima

What we must show. For a nonempty bounded $S \subseteq \mathbb{R}$ and $c > 0$, prove $\sup(cS) = c \sup S$ and $\inf(cS) = c \inf S$.

Prove the supremum statement using both defining properties.

Write $M = \sup S$. Every $s \in S$ satisfies $s \le M$; multiplication by the positive number c gives $cs \le cM$. Thus cM is an upper bound for cS. To show it is the least one, let $\varepsilon > 0$. Since $M - \varepsilon/c < M$, it is not an upper bound for S, so there is $s \in S$ with $M - \varepsilon/c < s \le M$. Multiplying gives $cM - \varepsilon < cs \le cM$. Hence no number below cM can be an upper bound for cS, proving the desired equality.

Repeat with lower bounds.

If $m = \inf S$, then $m \le s$ implies $cm \le cs$, so cm is a lower bound for cS. Given $\varepsilon > 0$, the number $m + \varepsilon/c$ is not a lower bound for S. Thus some $s \in S$ satisfies $m \le s < m + \varepsilon/c$, and $cm \le cs < cm + \varepsilon$. No number greater than cm is a lower bound for cS.

Conclusion. Both scaling identities follow; positivity of c was needed to preserve the inequality directions.

33.3 Exercise 33.3: Step functions and postage

What we must show. Integrate a finite step function and the postage-stamp example.

(a) Work one subinterval at a time.

Suppose $f = c_j$ on (u_{j-1}, u_j) for the finite partition in the exercise. On the closed interval $[u_{j-1}, u_j]$, the function differs from the constant c_j at most at its two endpoints. Exercise 32.7 therefore proves integrability there and gives integral $c_j(u_j - u_{j-1})$. Interval additivity, applied successively to the finitely many intervals, proves

$$\int_a^b f(x)\,dx = \sum_{j=1}^{m} c_j(u_j - u_{j-1}).$$

There is no condition on the finitely many partition-point values, except that they are real numbers.

(b) Apply the formula to the postage function.

On the four open unit intervals between 0 and 4, the values of P are $A, A+B, A+2B, A+3B$. Each interval has length 1; the value at 4 does not change the integral. Consequently

$$\int_0^4 P(x)\,dx = A + (A+B) + (A+2B) + (A+3B) = 4A + 6B.$$

Conclusion. Step functions are integrable, with integral equal to the sum of height times width; the postage integral is $4A + 6B$.

33.4 Exercise 33.4: An integrable absolute value does not suffice

What we must show. Find a nonintegrable f on $[0, 1]$ with integrable $|f|$. Set $f(x) = 1$ at rational points and $f(x) = -1$ at irrational points. Every nondegenerate interval contains both sorts of points. Thus every upper sum is 1 and every lower sum is -1; the two Darboux integrals differ, so f is not integrable. However, $|f(x)| = 1$ for every x, and a constant function is integrable, with integral 1 on $[0, 1]$.

Conclusion. This example shows that the converse of integrability of f implying integrability of $|f|$ is false.

33.5 Exercise 33.5: Bounding an integral without evaluating it

What we must show. Bound the absolute integral of $x^2 \sin^8(e^x)$ on $[-2\pi, 2\pi]$. The integrand is continuous, hence integrable. Since $0 \le \sin^8(e^x) \le 1$, we have $|x^2 \sin^8(e^x)| \le x^2$. The absolute-integral inequality and comparison give

$$\left|\int_{-2\pi}^{2\pi} x^2 \sin^8(e^x)\,dx\right| \le \int_{-2\pi}^{2\pi} x^2\,dx = 2\frac{(2\pi)^3}{3} = \frac{16\pi^3}{3}.$$

The integral of the square follows from the Darboux calculation in Section 32 (or, once established, the Fundamental Theorem).

Conclusion. The requested bound is $16\pi^3/3$.

33.6 Exercise 33.6: Absolute value does not increase oscillation

What we must show. Show, for a nonempty set S on which f is bounded,

$$\sup_S |f| - \inf_S |f| \le \sup_S f - \inf_S f.$$

Compare two values before taking extrema.

For arbitrary $x, y \in S$, the reverse triangle inequality gives

$$|f(x)| - |f(y)| \le |f(x) - f(y)| \le M - m, \quad M = \sup_S f, \quad m = \inf_S f.$$

The last inequality holds because both $f(x)$ and $f(y)$ lie in $[m, M]$. Fix y and take the supremum over x. This yields $\sup_S |f| \le |f(y)| + M - m$ for every $y \in S$. Taking the infimum over y gives $\sup_S |f| \le \inf_S |f| + M - m$. No maximum or minimum needs to be attained.

Conclusion. Rearrangement gives exactly the oscillation inequality needed in the proof of Theorem 33.5.

33.7 Exercise 33.7: Squaring preserves integrability

What we must show. Prove the Darboux-gap estimate and deduce integrability of f^2. Assume $|f(x)| \le B$ on $[a, b]$, with $B > 0$.

(a) Estimate the oscillation on one cell.

For any x, y in a partition cell J,

$$\left|f(x)^2 - f(y)^2\right| = |f(x) + f(y)|\,|f(x) - f(y)| \le 2B(\sup_J f - \inf_J f).$$

Taking the supremum over pairs $x, y \in J$ gives

$$\sup_J f^2 - \inf_J f^2 \le 2B(\sup_J f - \inf_J f).$$

Indeed, the difference of a supremum and an infimum equals the supremum of all differences; the values need not be attained. Multiply by the length of J and sum over the partition to obtain

$$U(f^2, P) - L(f^2, P) \le 2B[U(f, P) - L(f, P)].$$

(b) Use the integrability criterion.

If f is integrable, choose P with $U(f, P) - L(f, P) < \varepsilon/(2B)$. The estimate makes the gap for f^2 smaller than ε. Also f^2 is bounded by B^2, as required by the criterion.

Conclusion. f^2 is integrable whenever f is integrable.

33.8 Exercise 33.8: Products, maxima, and minima

What we must show. Prove integrability of three combinations of integrable f, g.

(a) Turn a product into a linear combination of squares.

Linearity already proves that $f + g$ and $f - g$ are integrable. By Exercise 33.7 their squares are integrable. Since

$$fg = \frac{(f + g)^2 - (f - g)^2}{4},$$

linearity proves integrability of fg. This argument establishes the existence of the integral, without claiming that an integral of a product is a product of integrals.

(b) Express maxima and minima using absolute value.

For any two real numbers u, v, $\max(u, v) = (u + v + |u - v|)/2$ and $\min(u, v) = (u + v - |u - v|)/2$. Check these identities separately when $u \geq v$ and $u < v$. Substituting $f(x), g(x)$ gives pointwise formulas. The absolute value of the integrable function $f - g$ is integrable by Theorem 33.5, and linearity now proves both conclusions.

Conclusion. The functions fg, $\max(f, g)$, and $\min(f, g)$ are integrable.

33.9 Exercise 33.9: Uniform limits of integrable functions

What we must show. Show that a uniform limit f is integrable and its integral is the limit of the integrals.

First check boundedness.

Choose one index m such that $|f - f_m| < 1$ everywhere. Since f_m is bounded, this also bounds f.

Make a Darboux gap small.

Given $\varepsilon > 0$, choose m with $\sup_{[a,b]} |f - f_m| < \varepsilon/[4(b-a)]$. Call this upper bound η. For every cell J,

$$\sup_J f \leq \sup_J f_m + \eta, \qquad \inf_J f \geq \inf_J f_m - \eta.$$

Since f_m is integrable, choose P with its gap below $\varepsilon/2$. Then

$$U(f, P) - L(f, P) \leq U(f_m, P) - L(f_m, P) + 2\eta(b - a) < \varepsilon.$$

The Darboux criterion proves integrability of f.

Estimate the difference of integrals.

Now $f - f_n$ is integrable, and

$$\left| \int_a^b f - \int_a^b f_n \right| \leq \int_a^b |f - f_n| \leq (b - a) \sup_{[a,b]} |f - f_n| \longrightarrow 0.$$

Uniform convergence is what makes the final supremum tend to zero. Theorem 25.2 treated continuous functions; this proof needs only integrability. The dominated-convergence result has weaker convergence hypotheses but additional boundedness and limit-integrability assumptions.

Conclusion. Uniform limits preserve Riemann integrability and allow the limit to pass through the integral.

33.10 Exercise 33.10: An oscillatory function that is still integrable

What we must show. Prove integrability on $[-1, 1]$ of $f(x) = \sin(1/x)$ for $x \neq 0$, with $f(0) = 0$.

Plan. Isolate the only troublesome point in a short interval. Given $\varepsilon > 0$, choose $0 < \delta < 1$ with $4\delta < \varepsilon/2$. The function is continuous on $[-1, -\delta]$ and $[\delta, 1]$, so those intervals have partitions whose

combined Darboux gaps are less than $\varepsilon/2$. Combine these partitions with the middle cell $[-\delta, \delta]$. On that cell $-1 \le f \le 1$, so its gap contribution is at most $2(2\delta) = 4\delta < \varepsilon/2$. The complete partition therefore has gap below ε.

Conclusion. The function is Riemann integrable. Infinitely many oscillations near one point can be confined to intervals of arbitrarily small length.

33.11 Exercise 33.11: An integrable function beyond the piecewise criteria

What we must show. Study $f(x) = x\,\mathrm{sgn}(\sin(1/x))$ for $x \neq 0$ and $f(0) = 0$, with $\mathrm{sgn}(0) = 0$.

(a) Rule out piecewise continuity.

At $x_k = 1/(k\pi) > 0$, the sine changes sign and the two one-sided limits of f are x_k and $-x_k$, in one order or the other. They differ because $x_k > 0$, so each x_k is a discontinuity. Every interval $(0, \delta)$ contains infinitely many of these points. A finite partition always leaves an interval $(0, \delta)$ without any interior partition point, and continuity on its interior is impossible. Thus f is not piecewise continuous on $[-1, 1]$.

(b) Rule out piecewise monotonicity.

The sign of $\sin(1/x)$ alternates on the infinitely many successive intervals between the x_k. In any $(0, \delta)$ choose ordered points $u < v < w$ in three consecutive sign intervals with signs $+, -, +$. Then $f(u) > 0$, $f(v) < 0$, and $f(w) > 0$. These values are neither nondecreasing nor nonincreasing. Hence f is not monotonic on any $(0, \delta)$, which again rules out any finite monotonicity partition.

(c) Estimate the Darboux gap directly.

Given $\varepsilon > 0$, choose $\delta \in (0, 1)$ with $4\delta^2 < \varepsilon/2$. On each of $[-1, -\delta]$ and $[\delta, 1]$ there are only finitely many zeros of $\sin(1/x)$. Between them the function equals x or $-x$; it is thus a finite piecewise continuous function, hence integrable there. Choose outer partitions with combined gaps below $\varepsilon/2$. On $[-\delta, \delta]$, $|f| \le |x| \le \delta$, so the gap contribution is at most $(2\delta)(2\delta) = 4\delta^2$. The combined gap is below ε.

Conclusion. f is neither piecewise continuous nor piecewise monotonic, but it is Riemann integrable. The two piecewise conditions are sufficient, not necessary.

33.12 Exercise 33.12: The rational-denominator function has zero integral

What we must show. For the function of Exercise 17.14, prove the two non-piecewise claims and compute the integral on every $[a, b]$. Recall that $f = 0$ at irrationals and $f(p/q) = 1/q$ for rationals in lowest terms with $q > 0$, including $f(0) = 1$.

(a) Examine every nondegenerate subinterval.

The function is discontinuous at every rational point: its value there is positive, whereas nearby irrational values are zero. Every nondegenerate interval contains a rational, so it cannot be an interval of continuity. Thus no finite continuity partition exists. It cannot be monotonic on any such interval either: choose a rational r strictly inside it and irrational points $u < r < v$ in it. Then $f(u) = f(v) = 0 < f(r)$, violating both possible monotonicities. Thus a finite monotonicity partition is also impossible.

(b) Isolate the finitely many large values.

Every cell contains an irrational, so every lower sum is zero. To make an upper sum small, let $\varepsilon > 0$ and choose N with $(b-a)/N < \varepsilon/2$. There are only finitely many rationals in $[a, b]$ with denominator $q \le N$: for each fixed q, only finitely many integers p have $a \le p/q \le b$. Make a partition whose cells meeting these finitely many points have total length less than $\varepsilon/2$. On these cells, use the bound $f \le 1$. On all other cells, all nonzero values have denominator $q > N$, and hence $f \le 1/N$. Therefore

$$U(f, P) \le 1 \cdot \frac{\varepsilon}{2} + \frac{1}{N}(b-a) < \varepsilon.$$

This proves the upper integral is also zero.

Conclusion. f is integrable on every bounded closed interval and $\int_a^b f = 0$, although it is neither of the two piecewise types.

33.13 Exercise 33.13: Equal integrals force an interior intersection

What we must show. For continuous f, g with equal integrals on $[a, b]$, find $x \in (a, b)$ with $f(x) = g(x)$. Let $h = f - g$. It is continuous and $\int_a^b h = 0$. Suppose h never vanishes in (a, b). By the Intermediate Value theorem it cannot take both a positive and a negative value there, since that would force a zero between them. Thus it is either positive everywhere in (a, b) or negative everywhere there. In the positive case, continuity gives $h \ge 0$ at the endpoints as well. At any chosen interior point x_0, continuity gives a small interval of positive length on which $h \ge h(x_0)/2 > 0$. Its integral is strictly positive, and the rest is nonnegative, contradicting the zero integral. Apply the same reasoning to $-h$ in the negative case.

Conclusion. The assumption was impossible, so $f(x) = g(x)$ for some interior x.

33.14 Exercise 33.14: A weighted mean-value theorem

What we must show. Prove the weighted formula, identify the ordinary case, and test the sign hypothesis.

(a) Separate the zero-weight case.

Let $G = \int_a^b g \ge 0$. If $G = 0$, nonnegative continuous g must be identically zero: a positive value would, by continuity, contribute a positive integral on a short interval. Thus $\int_a^b fg = 0$ too, and every interior x satisfies the formula. If $G > 0$, put $c = G^{-1} \int_a^b fg$. We need $f(x) = c$ at an interior point. Suppose no such point exists. As in Exercise 33.13, continuity forces $f - c$ to have one strict sign throughout (a, b). Because $G > 0$, g is positive at some interior point t_0; otherwise it would vanish on (a, b) and hence also at both endpoints. Thus $(f - c)g$ has one weak sign throughout $[a, b]$ and the corresponding strict sign on a neighborhood of t_0. Its integral cannot be zero. But linearity gives

$$\int_a^b (f - c)g = \int_a^b fg - cG = 0,$$

a contradiction. Therefore an interior x has $f(x) = c$, and $\int_a^b fg = f(x) \int_a^b g$.

(b) Recover the ordinary integral mean-value theorem.

Take $g \equiv 1$, so $G = b - a$. The formula becomes $f(x) = (b-a)^{-1} \int_a^b f$, exactly Theorem 33.9.

(c) Test a weight that changes sign.

On $[-1,1]$, set $f(t)=g(t)=t$. Then $\int_{-1}^{1} fg=\int_{-1}^{1} t^2\,dt=2/3$, but $f(x)\int_{-1}^{1} g=x\cdot 0=0$ for every x.

Conclusion. The weighted theorem holds for nonnegative continuous weights; part (c) shows that dropping the sign condition can destroy it.

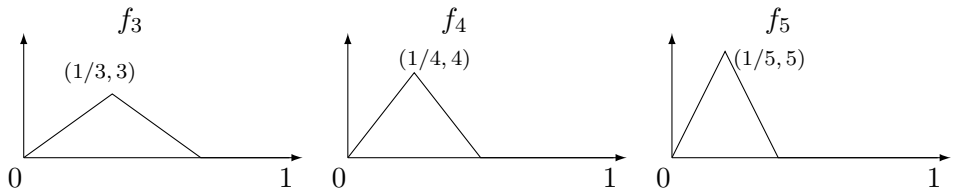
33.15 Exercise 33.15: Narrow triangular peaks

What we must show. Sketch the first three peaks, find their pointwise limit, and compare limits of integrals. The line-segment description gives the convenient formula

$$f_n(x)=\begin{cases} n^2x, & 0\le x\le 1/n,\\ 2n-n^2x, & 1/n\le x\le 2/n,\\ 0, & 2/n\le x\le 1. \end{cases}$$

(a) Draw the vertices and join them by straight segments.

The following sketches use the same horizontal and vertical scales; the vertical unit is 8 picture units and the horizontal unit is 100. The vertices are $(0,0),(1/n,n),(2/n,0),(1,0)$.



(b) Fix the point before choosing how large n must be.

At $x=0$, every value is zero. For a fixed $x>0$, choose $N>2/x$. If $n\ge N$, then $2/n<x$ and the last branch of the formula gives $f_n(x)=0$. Thus $f_n(x)\to 0$ for every $x\in[0,1]$.

(c) Compute areas and check the theorem's hypothesis.

Each graph is a triangle of base $2/n$ and height n, followed by a zero segment. Its integral is

$$\int_0^1 f_n(x)\,dx=\frac{1}{2}\frac{2}{n}\,n=1.$$

Hence the limit of the integrals is 1, whereas the integral of the pointwise limit is 0. There is no contradiction with Ross's Dominated Convergence theorem: it requires a single constant M bounding $|f_n(x)|$ for all n,x, but $f_n(1/n)=n$ is unbounded as n varies.

Conclusion. Pointwise convergence to zero does not force the integrals to tend to zero; the increasingly tall peaks retain area 1.

34 Fundamental Theorem of Calculus

There are two complementary facts to keep straight. If f is continuous, then $H(x) = \int_a^x f(t)\,dt$ is differentiable and $H'(x) = f(x)$. If $G' = f$ and the relevant integrability hypotheses hold, then $\int_a^b f = G(b) - G(a)$. An accumulation integral can remain continuous even at a jump of its integrand, although it may then have a corner. For reversed endpoints we use $\int_b^a f = -\int_a^b f$.

34.1 Exercise 34.1: Recovering the antiderivative evaluation formula

What we must show. Deduce $\int_a^b g' = g(b) - g(a)$ when g' is continuous. Define $F(x) = \int_a^x g'(t)\,dt$ for $x \in [a, b]$. Continuity of g' guarantees that it is integrable and, by Theorem 34.3, $F'(x) = g'(x)$ for interior x. Thus $(F - g)' = 0$ on (a, b). Both F and g are continuous on the closed interval, so Corollary 29.5 (the zero-derivative consequence of the Mean Value theorem) says $F - g$ is constant there. At a, $F(a) - g(a) = -g(a)$; hence $F(b) - g(b) = -g(a)$. Rearranging gives

$$\int_a^b g'(t)\,dt = F(b) = g(b) - g(a).$$

Conclusion. The continuous-integrand version of the Fundamental Theorem implies the antiderivative evaluation formula in this case.

34.2 Exercise 34.2: Two local averages of an exponential

What we must show. Calculate the two limits using derivatives of accumulation integrals.

(a) Center the integral at zero.

Let $H(x) = \int_0^x e^{t^2}\,dt$. The integrand is continuous, so $H'(0) = e^0 = 1$ and $H(0) = 0$. By the definition of derivative,

$$\lim_{x\to 0} \frac{1}{x} \int_0^x e^{t^2}\,dt = \lim_{x\to 0} \frac{H(x) - H(0)}{x - 0} = H'(0) = 1.$$

The signed-integral convention makes this valid from both sides.

(b) Center the integral at three.

Let $K(y) = \int_3^y e^{t^2}\,dt$. Then $K(3) = 0$ and $K'(3) = e^9$. Therefore

$$\lim_{h\to 0} \frac{1}{h} \int_3^{3+h} e^{t^2}\,dt = \lim_{h\to 0} \frac{K(3 + h) - K(3)}{h} = e^9.$$

Conclusion. The limits are 1 and e^9, respectively.

34.3 Exercise 34.3: A continuous accumulation function with one corner

What we must show. Integrate the piecewise function, sketch its accumulation function, and locate all derivatives. The integrand is 0 for $t < 0$, t for $0 \le t \le 1$, and 4 for $t > 1$.

(a) Integrate over the pieces actually traversed.

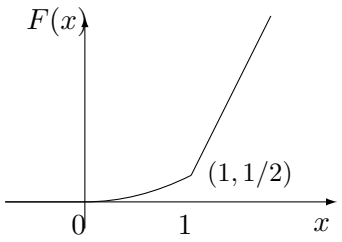
For $x \le 0$, $F(x) = \int_0^x 0\,dt = 0$. For $0 \le x \le 1$, $F(x) = \int_0^x t\,dt = x^2/2$. For $x \ge 1$, split at 1:

$$F(x) = \int_0^1 t\,dt + \int_1^x 4\,dt = \tfrac{1}{2} + 4(x-1) = 4x - \tfrac{7}{2}.$$

The isolated value of the integrand at 1 has no effect on the latter integral.

(b) Join the graph pieces at their actual heights.

The graph is horizontal at height zero to the left of 0, then follows a parabola to $(1, 1/2)$, then a straight line of slope 4. At 0 both neighboring formulas give 0; at 1 both give $1/2$. The formulas are continuous on their separate pieces, so F is continuous at every real number.



The drawing scales the horizontal coordinate by 40 and the vertical coordinate by 20, so the right line has its correct slope in data coordinates.

(c) Check the joining points separately.

Away from $0, 1$, differentiation gives $0, x, 4$ on the three open pieces. At zero, the difference quotient is 0 for negative increments and $h/2$ for positive increments, so $F'(0) = 0$. At 1 the left-hand difference quotient tends to 1 and the right-hand one tends to 4. Thus no derivative exists there.

Conclusion. F is continuous on $\mathbb{R}$ and differentiable precisely on $\mathbb{R} \setminus \{1\}$, with $F' = 0$ for $x \le 0$, $F' = x$ for $0 < x < 1$, and $F' = 4$ for $x > 1$.

34.4 Exercise 34.4: Two corners in another accumulation function

What we must show. Repeat the analysis when $f(t) = t$ for $t < 0$, $f(t) = t^2 + 1$ for $0 \le t \le 2$, and $f(t) = 0$ for $t > 2$.

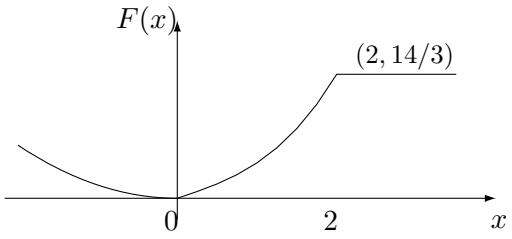
(a) Keep the orientation of the integral for negative x.

For $x < 0$, $F(x) = -\int_x^0 t\,dt = x^2/2$. For $0 \le x \le 2$, integration gives $F(x) = x^3/3 + x$. For $x > 2$, the integrand vanishes after 2, so $F(x) = F(2) = 8/3 + 2 = 14/3$. Thus

$$F(x) = \begin{cases} x^2/2, & x \le 0, \\ x^3/3 + x, & 0 \le x \le 2, \\ 14/3, & x \ge 2. \end{cases}$$

(b) Sketch and test continuity.

The left parabola decreases to $(0, 0)$; the middle cubic expression increases to $(2, 14/3)$; the remaining graph is horizontal. The two formulas at zero both give zero, and those at 2 both give $14/3$. Thus the function is continuous everywhere.



This is a schematic plot of the exact formulas; the middle curve is represented by closely spaced points.

(c) Compare slopes at the joins.

On the three open pieces the derivative is respectively x, x^2+1, and 0. At 0, the one-sided derivatives are 0 and 1. At 2, they are 5 and 0. The unequal one-sided derivatives exclude differentiability at each joining point.

Conclusion. F is continuous on all of $\mathbb{R}$, differentiable on $\mathbb{R}\setminus\{0,2\}$, and has the piecewise derivative just computed.

34.5 Exercise 34.5: Moving both endpoints

What we must show. Differentiate $F(x)=\int_{x-1}^{x+1} f(t)\,dt$ for continuous f on $\mathbb{R}$. Choose the fixed-base primitive $H(u)=\int_0^u f(t)\,dt$. The Fundamental Theorem gives $H'(u)=f(u)$ for every u. Interval additivity, with signed integrals when necessary, gives

$$F(x)=H(x+1)-H(x-1).$$

Apply the chain rule to each term. The derivatives of $x+1$ and $x-1$ are both 1, so

$$F'(x)=H'(x+1)-H'(x-1)=f(x+1)-f(x-1).$$

Conclusion. F is differentiable everywhere, with the displayed derivative.

34.6 Exercise 34.6: A trigonometric upper endpoint

What we must show. Differentiate $G(x)=\int_0^{\sin x} f(t)\,dt$ for continuous f. As above, let $H(u)=\int_0^u f(t)\,dt$, so $H'=f$ everywhere. Now $G=H\circ\sin$, and the chain rule yields

$$G'(x)=H'(\sin x)\cos x=f(\sin x)\cos x.$$

This works even when $\sin x<0$ because H uses oriented integrals, and at zeros of $\cos x$ because the chain rule includes zero inner derivatives.

Conclusion. G is differentiable on all of $\mathbb{R}$ and $G'(x)=f(\sin x)\cos x$.

34.7 Exercise 34.7: A substitution with a reversed direction

What we must show. Evaluate $\int_0^1 x\sqrt{1-x^2}\,dx$ by change of variables. Choose $u=1-x^2$, so $du=-2x\,dx$ and $x\,dx=-du/2$. As x goes from 0 to 1, u goes from 1 to 0. Keeping these transformed

endpoints gives

$$\int_0^1 x\sqrt{1-x^2}\,dx = -\frac{1}{2}\int_1^0 u^{1/2}\,du = \frac{1}{2}\int_0^1 u^{1/2}\,du = \frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_0^1 = \frac{1}{3}.$$

The substitution is justified by the change-of-variables theorem: its inner map is continuously differentiable and $\sqrt{u}$ is continuous on the transformed interval $[0,1]$. No derivative of $\sqrt{u}$ at zero is needed.

Conclusion. The value of the integral is $1/3$.

34.8 Exercise 34.8: Two choices of antiderivative in integration by parts

What we must show. Evaluate $I = \int_0^1 x\arctan x\,dx$ using two versions of v.

(a) Choose $u = \arctan x$ and $v = x^2/2$.

Then $u' = 1/(1+x^2)$ and $v' = x$. Integration by parts gives

$$I = \left[\frac{x^2}{2}\arctan x\right]_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{1+x^2}\,dx.$$

The boundary term is $\pi/8$. Since $x^2/(1+x^2) = 1 - 1/(1+x^2)$, the remaining integral is $1-\pi/4$. Therefore

$$I = \frac{\pi}{8} - \frac{1}{2}\left(1 - \frac{\pi}{4}\right) = \frac{\pi}{4} - \frac{1}{2}.$$

(b) Choose instead $v = (x^2+1)/2$.

This still has derivative x. Now the denominator cancels immediately:

$$I = \left[\frac{x^2+1}{2}\arctan x\right]_0^1 - \int_0^1 \frac{x^2+1}{2(1+x^2)}\,dx = \frac{\pi}{4} - \frac{1}{2}.$$

Both calculations are valid because all functions and derivatives used are continuous on $[0,1]$.

Conclusion. The common answer is $\pi/4 - 1/2$. A useful constant in v can simplify the remaining integral.

34.9 Exercise 34.9: Integrating the inverse sine

What we must show. Evaluate $\int_0^{1/2} \arcsin x\,dx$. The inverse-function integration identity of Example 3, or integration by parts with $u = \arcsin x$ and $v = x$, gives

$$\int_0^{1/2} \arcsin x\,dx = [x\arcsin x]_0^{1/2} - \int_0^{1/2} \frac{x}{\sqrt{1-x^2}}\,dx.$$

Here $\arcsin(1/2) = \pi/6$, so the boundary term is $\pi/12$. Since $d(\sqrt{1-x^2})/dx = -x/\sqrt{1-x^2}$,

$$\int_0^{1/2} \frac{x}{\sqrt{1-x^2}}\,dx = \left[-\sqrt{1-x^2}\right]_0^{1/2} = 1 - \frac{\sqrt{3}}{2}.$$

All derivatives in this calculation are continuous on $[0,1/2]$, which stays away from the singular endpoints ± 1.

Conclusion. $\displaystyle\int_0^{1/2} \arcsin x\,dx = \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1.$

34.10 Exercise 34.10: The areas under a function and its inverse

What we must show. Give a geometric proof that $\int_0^1 g(x)\,dx + \int_0^1 g^{-1}(u)\,du = 1$. Because g is a continuous strictly increasing bijection of $[0,1]$ onto itself, $g(0) = 0$, $g(1) = 1$, and its inverse is continuous. Consider the unit square with horizontal coordinate x and vertical coordinate u. The graph $u = g(x)$ divides it into two regions. The region below the graph has area $\int_0^1 g(x)\,dx$, by vertical slices. The complementary region is described by $g(x) \le u$; strict increase makes this equivalent to $x \le g^{-1}(u)$. Its horizontal slice at height u therefore has length $g^{-1}(u)$, so its area is $\int_0^1 g^{-1}(u)\,du$. Together the two regions fill the square, sharing only their boundary curve, so their areas sum to its area 1.

Conclusion. The two integrals are the complementary areas in the unit square.

34.11 Exercise 34.11: A continuous square with zero integral

What we must show. Show that $\int_a^b f(x)^2\,dx = 0$ forces continuous f to vanish everywhere. Suppose instead that $f(c) \neq 0$ for some $c \in [a,b]$. Continuity at c gives a relative neighborhood on which $|f(x) - f(c)| < |f(c)|/2$. The reverse triangle inequality gives $|f(x)| \ge |f(c)| - |f(x) - f(c)| > |f(c)|/2$ there. Even if c is an endpoint, this relative neighborhood contains a closed subinterval $J \subseteq [a,b]$ of positive length. Thus

$$\int_a^b f(x)^2\,dx \ge \int_J f(x)^2\,dx \ge \frac{|f(c)|^2}{4}\,|J| > 0,$$

a contradiction. The comparison is justified because f^2 is continuous and nonnegative throughout the interval.

Conclusion. No such c exists, so $f(x) = 0$ for every $x \in [a,b]$.

34.12 Exercise 34.12: Testing against every continuous function

What we must show. If $\int_a^b f(x)g(x)\,dx = 0$ for every continuous g, prove that the continuous function f is identically zero. The assumption permits *any* continuous test function g; in particular, choose $g = f$, which is continuous by hypothesis. The assumed identity becomes

$$\int_a^b f(x)^2\,dx = 0.$$

Exercise 34.11 applies directly and yields $f = 0$ everywhere. Conversely, the zero function clearly has zero integral against every continuous test function, although only the forward implication was requested.

Conclusion. f must vanish identically on $[a,b]$.

35 Riemann–Stieltjes integrals

Read this before calculating. An ordinary integral weights an interval by its length. A Stieltjes integral weights it by the increase of an increasing function F. Here "increasing" permits equality: $s < t$ implies $F(s) \le F(t)$. Ross uses a modified definition that handles jumps explicitly. This matters: a formula from another book's traditional Stieltjes theory may not apply to this section.

Write $F(t-)$ and $F(t+)$ for the one-sided limits. On the particular interval $[a, b]$ being integrated over, the endpoint conventions are $F(a-) = F(a)$ and $F(b+) = F(b)$. Thus the mass at t is $\Delta F(t) = F(t+) - F(t-)$. For a partition $P = \{a = t_0 < \cdots < t_n = b\}$ set

$$w_k = F(t_k-) - F(t_{k-1}+), \qquad J_F(f, P) = \sum_{k=0}^{n} f(t_k)\Delta F(t_k).$$

The upper sum is $J_F(f, P) + \sum_{k=1}^{n} (\sup_{(t_{k-1}, t_k)} f) w_k$; the lower sum uses infima instead. The numbers w_k and $\Delta F(t_k)$ are nonnegative, and together they add to $W = F(b) - F(a)$. In particular, if $|h| \le B$, then every upper and lower sum for h lies between $-BW$ and BW.

The essential test is Theorem 35.6: a bounded function is F-integrable precisely when, for every $\varepsilon > 0$, some partition satisfies $U_F(f, P) - L_F(f, P) < \varepsilon$. The jump terms cancel in this difference. We will often use the equivalent expression

$$U_F(f, P) - L_F(f, P) = \sum_{k=1}^{n} \operatorname{osc}(f, (t_{k-1}, t_k)) w_k,$$

where $\operatorname{osc}(f, I) = \sup_I f - \inf_I f$ is the oscillation of f on I.

35.1 Exercise 35.1: Why an increasing function has one-sided limits

What we must show. Identify both one-sided limits of F by a supremum or an infimum, including the indicated endpoints.

Plan. Increasing values cannot move back and forth. Their supremum on the left, or infimum on the right, is therefore the only possible limiting value. We verify this using the definition of a limit.

(a) Approach $t \in (a, b]$ from the left.

The set $A = \{F(x) : a < x < t\}$ is nonempty and bounded above by $F(t)$. Consequently $L = \sup A$ is a finite real number. Given $\varepsilon > 0$, the definition of supremum supplies $y \in (a, t)$ such that $L - \varepsilon < F(y) \le L$. For every x with $y < x < t$, monotonicity gives

$$L - \varepsilon < F(y) \le F(x) \le L.$$

Take $\delta = t - y > 0$. If $x \in [a, b]$ and $0 < t - x < \delta$, then $y < x < t$, so $|F(x) - L| < \varepsilon$. This is exactly $\lim_{x \to t-} F(x) = L$. The proof works when $t = b$, since it only uses points to the left of t.

(b) Approach $t \in [a, b)$ from the right.

The nonempty set $B = \{F(x) : t < x < b\}$ is bounded below by $F(t)$. Put $R = \inf B$. Given $\varepsilon > 0$, choose $y \in (t, b)$ with $R \le F(y) < R + \varepsilon$. Whenever $t < x < y$,

$$R \le F(x) \le F(y) < R + \varepsilon.$$

Now $\delta = y - t$ proves $\lim_{x \to t+} F(x) = R$. At $t = a$ the same argument is valid using only points to the right.

Conclusion. The left limit is $\sup\{F(x) : a < x < t\}$ and the right limit is $\inf\{F(x) : t < x < b\}$. Both are finite.

35.2 Exercise 35.2: Separate a continuous increase from a jump

What we must show. Evaluate $\int_0^3 x^2\, dF(x)$ for the piecewise function in Example 4.

Record all three pieces, especially the jump.

The function is $F(t) = 0$ for $t < 0$, $F(t) = t^2$ for $0 \le t < 2$, and $F(t) = t + 5$ for $t \ge 2$. At 2 its left limit is 4 and its right limit is 7, so the jump has size 3. Write $F = F_1 + 3J_2$, where on $[0, 3]$

$$F_1(t) = \begin{cases} t^2, & 0 \le t \le 2, \\ t + 2, & 2 \le t \le 3. \end{cases}$$

This F_1 is continuous. Its derivative is $2t$ on the first piece and 1 on the second. We apply Theorem 35.13 separately on the two pieces; no derivative at the joining point is needed.

Calculate each contribution.

Linearity in the integrator, interval additivity, and the jump rule $\int f\, dJ_2 = f(2)$ give

$$\begin{aligned} \int_0^3 x^2\, dF(x) &= \int_0^2 x^2(2x)\, dx + \int_2^3 x^2\, dx + 3(2^2) \\ &= \left[\frac{x^4}{2}\right]_0^2 + \left[\frac{x^3}{3}\right]_2^3 + 12 \\ &= 8 + \frac{27 - 8}{3} + 12 = \frac{79}{3}. \end{aligned}$$

The extra 12 is essential: differentiating the smooth pieces alone would lose the mass concentrated at 2.

Conclusion. $\displaystyle\int_0^3 x^2\, dF(x) = 79/3.$

35.3 Exercise 35.3: Integrating against the integer step function

What we must show. Evaluate the three requested integrals against $F(t) = \lfloor t \rfloor$.

Plan. This function has jumps of size 1 at the integers. On a specified integration interval we include only the jumps belonging to the restricted function, with Ross's endpoint convention.

(a) Integrate x on $[0, 6]$.

On this interval $F(0-)$ means $F(0) = 0$, rather than the limit of the original function from negative numbers. Also $F(0+) = 0$. There is therefore no mass at 0. There is a mass of 1 at each of $1, 2, 3, 4, 5, 6$. The last one is included because $F(6+) = F(6) = 6$ and $F(6-) = 5$. Example 1 gives

$$\int_0^6 x\, dF(x) = \sum_{j=1}^{6} j = \frac{6 \cdot 7}{2} = 21.$$

(b) Integrate x^2 on $[0, 3]$.

The contributing jumps are at $1, 2, 3$, with the same endpoint reasoning. Thus

$$\int_0^3 x^2 \, dF(x) = 1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14.$$

(c) Integrate x^2 on $[1/4, \pi/4]$.

Both endpoints lie strictly between 0 and 1. The integrator is identically 0 on the whole interval; every interval increment and every endpoint jump is zero. Its upper and lower sums are both zero.

Conclusion. The three values are 21, 14, and 0, respectively.

35.4 Exercise 35.4: A smooth trigonometric integrator

What we must show. Integrate x against $F(t) = \sin t$ on the two specified intervals.

Check the conversion to an ordinary integral.

On $[-\pi/2, \pi/2]$, $F'(t) = \cos t \geq 0$, so F is increasing. Both F' and $f(t) = t$ are continuous. Theorem 35.13 therefore permits the replacement $dF(x) = \cos x \, dx$ on each interval.

(a) The interval $[0, \pi/2]$.

Integration by parts with $u = x$ and $dv = \cos x \, dx$ gives

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx = x \sin x + \cos x.$$

Consequently

$$\int_0^{\pi/2} x \, dF(x) = [x \sin x + \cos x]_0^{\pi/2} = \frac{\pi}{2} - 1.$$

(b) The interval $[-\pi/2, \pi/2]$.

The function $x \cos x$ is odd: replacing x by $-x$ changes its sign. Its integrals over the two halves cancel, so the answer is 0. Alternatively the displayed antiderivative has value $\pi/2$ at both endpoints, giving difference 0.

Conclusion. The values are $\pi/2 - 1$ and 0.

35.5 Exercise 35.5: The rational indicator sees continuous and atomic mass differently

What we must show. For $f = \mathbf{1}_{\mathbb{Q}}$, contrast continuous integrators with the purely discrete integrators in Examples 1 and 3.

(a) A continuous nonconstant integrator.

Every nonempty open interval contains both a rational number and an irrational number. Hence on every partition interval (t_{k-1}, t_k), the supremum of f is 1 and the infimum is 0. Continuity of F

makes all jump terms zero and gives $w_k = F(t_k) - F(t_{k-1})$. Therefore, for every partition,

$$L_F(f, P) = 0, \qquad U_F(f, P) = \sum_{k=1}^{n} [F(t_k) - F(t_{k-1})] = F(b) - F(a).$$

Taking the supremum of the lower sums and infimum of the upper sums changes neither number. Since $F(b) > F(a)$, the lower and upper integrals are different. Thus f is not F-integrable.

(b) A finite or summable collection of point masses.

For $F = \sum_j c_j J_{u_j}$ as in those examples, every bounded function is F-integrable, including this f. Its integral is

$$\int_a^b f \, dF = \sum_j c_j f(u_j) = \sum_{\{j : u_j \in \mathbb{Q}\}} c_j.$$

The formula includes the finite case. In the infinite case it converges absolutely, because $0 \le c_j f(u_j) \le c_j$ and $\sum_j c_j < \infty$. These integrators assign their mass to actual points; the value of f at each mass point is well defined.

Conclusion. For continuous F of positive total increase, f is not integrable. For the discrete F of Examples 1 and 3, it is integrable, with integral equal to the total mass at rational points.

35.6 Exercise 35.6: Uniform limits preserve Stieltjes integrability

What we must show. If each f_n is F-integrable and $f_n \to f$ uniformly on $[a, b]$, prove that f is F-integrable and that the integrals converge.

Plan. Uniform convergence controls every point by the same error. Multiplying that error by the total mass $W = F(b) - F(a)$ controls every upper sum, lower sum, and ultimately the integral.

First establish boundedness.

Take one index N for which $|f(x) - f_N(x)| < 1$ for every $x \in [a, b]$. Such an index exists by uniform convergence. Since f_N is integrable, it is bounded, say $|f_N(x)| \le B$. Then $|f(x)| \le B + 1$ everywhere. This verifies the boundedness required before applying the Darboux criterion.

Make the upper–lower gap small.

Assume $W > 0$; if $W = 0$, every sum and integral is zero and the conclusions are immediate. Fix $\varepsilon > 0$ and select N so that $|f - f_N| \le \eta$ everywhere, where $\eta = \varepsilon/(4W)$. On any nonempty interval I,

$$\sup_I f \le \sup_I f_N + \eta, \qquad \inf_I f \ge \inf_I f_N - \eta.$$

Subtracting gives $\operatorname{osc}(f, I) \le \operatorname{osc}(f_N, I) + 2\eta$. Because f_N is integrable, choose P with $U_F(f_N, P) - L_F(f_N, P) < \varepsilon/2$. Since $\sum_k w_k \le W$,

$$\begin{aligned} U_F(f, P) - L_F(f, P) &\le U_F(f_N, P) - L_F(f_N, P) + 2\eta \sum_k w_k \\ &< \frac{\varepsilon}{2} + 2\eta W = \varepsilon. \end{aligned}$$

Theorem 35.6 proves that f is integrable.

Now estimate the integrals.

Put $\eta_n = \sup_{x\in[a,b]} |f_n(x) - f(x)|$; uniform convergence says $\eta_n \to 0$. The integrable difference $f_n - f$ satisfies $-\eta_n \le f_n - f \le \eta_n$. Order preservation and integration of constants give

$$-\eta_n W \le \int_a^b (f_n - f)\, dF \le \eta_n W.$$

Thus $\left|\int_a^b f_n\, dF - \int_a^b f\, dF\right| \le \eta_n W \to 0$.

Conclusion. Uniform convergence permits passage to the limit under this integral, with error at most the uniform function error times the total mass.

35.7 Exercise 35.7: Products, squares, maxima, and minima

What we must show. Show that all four indicated constructions preserve F-integrability.

(a) Squaring an integrable function.

Choose $B \ge 1$ such that $|f(x)| \le B$ on $[a, b]$. For any two points x, y in the same interval I,

$$\left|f(x)^2 - f(y)^2\right| = |f(x) - f(y)|\,|f(x) + f(y)| \le 2B\,|f(x) - f(y)|\,.$$

The supremum of pairwise differences of a bounded real function is its oscillation. It follows that $\operatorname{osc}(f^2, I) \le 2B \operatorname{osc}(f, I)$. Given $\varepsilon > 0$, take a partition making the gap for f less than $\varepsilon/(2B)$. The gap formula then makes the gap for f^2 less than ε. Since f^2 is bounded, the criterion proves its integrability.

(b) Multiplying two integrable functions.

Linearity makes $f + g$ and $f - g$ integrable. Part (a) makes their squares integrable. The identity

$$fg = \frac{(f+g)^2 - (f-g)^2}{4}$$

then expresses fg as a linear combination of integrable functions. To check the identity, expand both squares: their f^2 and g^2 terms cancel, leaving $4fg$ in the numerator.

(c) Taking the pointwise maximum and minimum.

Theorem 35.10 makes $|f - g|$ integrable. At each point,

$$\max(f, g) = \frac{f + g + |f - g|}{2}, \qquad \min(f, g) = \frac{f + g - |f - g|}{2}.$$

For example, if $f(x) \ge g(x)$ then $|f(x) - g(x)| = f(x) - g(x)$, so the first expression becomes $f(x)$ and the second becomes $g(x)$. The other ordering gives the reversed roles. The formulas and linearity prove both integrability assertions.

Conclusion. f^2, fg, $\max(f, g)$, and $\min(f, g)$ are all F-integrable.

35.8 Exercise 35.8: A continuous density defines the weight

What we must show. For continuous $g \geq 0$ and $F(t) = \int_a^t g(x)\,dx$, show that $\int_a^b f\,dF = \int_a^b f(x)g(x)\,dx$ for continuous f.

Check that F is an admissible increasing integrator.

If $a \leq s < t \leq b$, interval additivity gives

$$F(t) - F(s) = \int_s^t g(x)\,dx \geq 0,$$

because g is nonnegative. Thus F is increasing. The Fundamental Theorem of Calculus, applied to continuous g, gives $F' = g$ on the interior, with the corresponding one-sided derivatives at the endpoints. In particular F' is continuous on the closed interval.

Apply the smooth-integrator formula.

All the hypotheses of Theorem 35.13 are now verified, so

$$\int_a^b f(x)\,dF(x) = \int_a^b f(x)F'(x)\,dx = \int_a^b f(x)g(x)\,dx.$$

If g is identically zero, F is constant. Although the section usually excludes a constant integrator to avoid trivial cases, its sums are still defined and all equal zero; the formula is then also true. If g is nonzero somewhere, continuity and nonnegativity give positive integral, so $F(b) > F(a)$ as in the standing convention.

Conclusion. The density g converts a Stieltjes integral into the usual weighted Riemann integral.

35.9 Exercise 35.9: A weighted mean value is attained

What we must show. Prove the mean-value assertion for continuous f, then recover the weighted integral theorem in Exercise 33.14.

(a) Trap the average between a minimum and a maximum.

The Extreme Value Theorem gives real numbers $m = \min_{[a,b]} f$ and $M = \max_{[a,b]} f$. Put $W = F(b) - F(a) > 0$. The continuous function f is F-integrable by Theorem 35.7. Integrating $m \leq f \leq M$ with order preservation gives

$$mW \leq \int_a^b f\,dF \leq MW.$$

Divide by the positive number W and call the quotient A. We have $m \leq A \leq M$. By the Intermediate Value Theorem, f takes every value between its attained minimum and maximum. Hence some $x \in [a, b]$ satisfies $f(x) = A$. Multiplication by W proves

$$\int_a^b f\,dF = f(x)[F(b) - F(a)].$$

The conclusion naturally allows endpoints: all the mass could be concentrated at a, for example.

(b) Recover the continuous-density theorem.

For continuous $g \geq 0$ set $F(t) = \int_a^t g(s)\,ds$, as in Exercise 35.8. If $g \equiv 0$, both integrals in Exercise 33.14(a) are zero and any $x \in (a, b)$ works. Otherwise $W = \int_a^b g > 0$. Part (a) and Exercise 35.8 give

$$\int_a^b f(t)g(t)\,dt = f(x)\int_a^b g(t)\,dt$$

initially for some $x \in [a, b]$.

Explain why the point can be chosen in the interior here.

Exercise 33.14(a) asks for $x \in (a, b)$, so this small extra point deserves a proof. Let $A = (\int_a^b fg)/(\int_a^b g)$. Suppose no interior point satisfies $f(x) = A$. By continuity and the Intermediate Value Theorem, $f - A$ must then have the same strict sign on all of (a, b). Also some $t_0 \in (a, b)$ has $g(t_0) > 0$: a positive endpoint value of g would imply positive nearby interior values as well. If $f - A > 0$, continuity gives a short interval around t_0 on which both $f - A$ and g have positive lower bounds. Thus $(f - A)g \geq 0$ everywhere and has strictly positive integral, contrary to

$$\int_a^b (f - A)g = \int_a^b fg - A\int_a^b g = 0.$$

If $f - A < 0$, apply the same argument to $(A - f)g$ and obtain the same contradiction. Therefore an interior point does attain A. Taking $g \equiv 1$ recovers the ordinary integral mean-value theorem, as requested in Exercise 33.14(b).

Conclusion. Part (a) is the Stieltjes mean-value formula. With a continuous nonnegative density it gives Exercise 33.14(a), including its interior-point conclusion.

35.10 Exercise 35.10: A polynomial integrated against itself

What we must show. For a positive odd integer n and $F(x) = f(x) = x^n$, evaluate $\int_a^b f\,dF$.

Explain the parity hypothesis.

The map $x \mapsto x^n$ is increasing on all of $\mathbb{R}$ when n is odd. Its derivative is $nx^{n-1} \geq 0$ since $n - 1$ is even. Both f and F' are continuous, so Theorem 35.13 applies on any closed interval $[a, b]$.

Convert and integrate.

We have

$$\begin{aligned}\int_a^b x^n\,d(x^n) = \int_a^b x^n\,nx^{n-1}\,dx &= n\int_a^b x^{2n-1}\,dx \\ &= n\left[\frac{x^{2n}}{2n}\right]_a^b = \frac{1}{2}(b^{2n} - a^{2n}).\end{aligned}$$

The result can be negative if the interval lies on the negative half-line. That is consistent: the weight is nonnegative, but $f(x) = x^n$ is negative there.

Conclusion. $\int_a^b f\,dF = \frac{1}{2}[b^{2n} - a^{2n}]$.

35.11 Exercise 35.11: Change variables without differentiating the substitution

What we must show. For a continuous strictly increasing map $\phi : [c, d] \to [a, b]$ with the stated endpoint values, prove integrability and equality of the two Stieltjes integrals for $g = f \circ \phi$ and $G = F \circ \phi$.

Plan. A partition and its image under ϕ describe exactly the same intervals and mass points. Match their sums term by term. There is no assumption that ϕ is differentiable.

The substitution maps intervals onto intervals.

Continuity, strict increase, and the endpoint values imply that ϕ is a bijection from $[c, d]$ onto $[a, b]$. Its inverse is continuous: for an interior u, points just below and above u have image values strictly below and above $\phi(u)$, which trap the inverse of every sufficiently nearby image point. The same one-sided trapping works at the endpoints. In particular,

$$\phi((u, v)) = (\phi(u), \phi(v)) \qquad (c \le u < v \le d).$$

The composition G is increasing, and g is bounded because f is bounded on the image $[a, b]$.

Match one-sided limits and jumps.

For $u \in (c, d)$, as $v \to u-$ the numbers $\phi(v)$ approach $\phi(u)$ from below. They cover a whole one-sided interval near $\phi(u)$. Therefore

$$G(u-) = F(\phi(u)-), \qquad G(u+) = F(\phi(u)+).$$

At $u = c, d$ the endpoint conventions give these same identities for the relevant decreed limits. Subtracting proves $\Delta G(u) = \Delta F(\phi(u))$.

Match the upper and lower sums.

Take any $Q = \{c = u_0 < \cdots < u_n = d\}$ and its image partition $P = \{a = \phi(u_0) < \cdots < \phi(u_n) = b\}$. Bijection on the open subintervals gives

$$\sup_{(u_{k-1}, u_k)} g = \sup_{(\phi(u_{k-1}), \phi(u_k))} f,$$

and the analogous identity for infima. The one-sided-limit identities give matching interval weights, and $g(u_k) = f(\phi(u_k))$ gives matching jump terms. Thus

$$U_G(g, Q) = U_F(f, P), \qquad L_G(g, Q) = L_F(f, P).$$

Every partition of $[a, b]$ arises this way, by applying ϕ^{-1} to its points. Consequently the infima of the upper sums agree, and the suprema of the lower sums agree. Since f is F-integrable, those two common numbers coincide.

Conclusion. g is G-integrable, and $\displaystyle\int_c^d g\, dG = \int_a^b f\, dF$.

35.12 Exercise 35.12: A strictly increasing function with dense jumps

What we must show. Analyze the sum of positive jumps placed at every rational point of $[a, b]$.

First address the wording in the supplied edition.

The printed exercise says that the c_j are "positive integers" and that $\sum_j c_j < \infty$. These conditions are incompatible: each partial sum is at least its number of terms, so the series would diverge to $+\infty$. Thus there is no example satisfying the literal hypotheses. The intended, nonvacuous version uses *positive real numbers* with finite sum, just as in Example 3. We solve that version explicitly; for instance $c_j = 2^{-j}$ is an allowed choice. Let (u_j) enumerate the rationals in $[a, b]$ without repetition, and retain Ross's special convention for J_a.

(a) Show that the sum exists and increases strictly.

Since $0 \le J_{u_j}(t) \le 1$,

$$0 \le \sum_{j=1}^{N} c_j J_{u_j}(t) \le \sum_{j=1}^{\infty} c_j < \infty.$$

So the nondecreasing partial sums have a finite limit $F(t)$ at every point. Also their tails are at most $\sum_{j>N} c_j$, independently of t, so the convergence is uniform. For $a \le s < t \le b$, density of $\mathbb{Q}$ supplies a rational $u_m \in (s, t)$. The function J_{u_m} is 0 at s and 1 at t. Every other summand is increasing, hence has nonnegative increment. Subtracting the convergent series gives

$$F(t) - F(s) = \sum_{j=1}^{\infty} c_j [J_{u_j}(t) - J_{u_j}(s)] \ge c_m > 0.$$

This proves strict increase, not merely nondecrease.

(b) Prove continuity at every irrational point.

Let $t \in [a, b] \setminus \mathbb{Q}$ and $\varepsilon > 0$. Choose N with $\sum_{j>N} c_j < \varepsilon$. Because t differs from each of the finitely many $u_1, \ldots, u_N$, there is $\delta > 0$ such that $|x - t| < \delta$ crosses none of these points. Each of the first N step functions has the same value at x and t. For the remaining terms, $\left|J_{u_j}(x) - J_{u_j}(t)\right| \le 1$. Thus for $x \in [a, b]$ with $|x - t| < \delta$,

$$|F(x) - F(t)| \le \sum_{j>N} c_j \left|J_{u_j}(x) - J_{u_j}(t)\right| \le \sum_{j>N} c_j < \varepsilon.$$

This proves relative continuity at t, including an irrational endpoint if one occurs.

Show discontinuity at every rational point.

If $u_m \in (a, b]$, then $J_{u_m}(u_m) = 1$, whereas $J_{u_m}(x) = 0$ for every $x < u_m$. Since every summand is increasing, $F(u_m) - F(x) \ge c_m$ for $x < u_m$ in $[a, b]$. Points of the domain can approach u_m from the left while this gap stays at least $c_m > 0$, so F is discontinuous there. If the rational point is $u_m = a$, Ross defines $J_a(a) = 0$ and $J_a(x) = 1$ for $x > a$. Thus $F(x) - F(a) \ge c_m$ for all $x > a$, which proves right discontinuity at a. In fact Example 3 shows that the jump at u_m is exactly c_m.

Conclusion. Under the intended summable positive-real-weight hypothesis, F is strictly increasing, continuous exactly at irrational points of $[a, b]$, and discontinuous at every rational point. Under the printed positive-integer hypothesis, no such weight sequence exists.

36 Improper integrals

An improper integral is a *limit of ordinary integrals*; it is not an instruction to substitute an infinite number into an antiderivative. For example,

$$\int_a^b f = \lim_{d\to b^-} \int_a^d f$$

when the left endpoint is harmless. If both endpoints are troublesome, choose an interior point α and take the two limits separately. A finite limit means convergence. A limit $+\infty$ or $-\infty$ is an extended value, not convergence. An oscillating limit has no value. In particular, a symmetric limit on $[-R, R]$ cannot replace the two one-sided limits in the definition.

For Stieltjes integrals, F is increasing. Its increment $F(v) - F(u)$ replaces the interval length $v - u$. We use Ross's endpoint convention from Section 35, so that adjacent integrals add without counting a jump twice. A distribution function has total increment $F(\infty) - F(-\infty) = 1$.

36.1 Exercise 36.1: Agreement with an ordinary integral

What we must show. If f is Riemann integrable on the finite interval $[a, b]$, prove that $\lim_{d\to b^-} \int_a^d f = \int_a^b f$.

Plan. The only omitted contribution is over the short interval $[d, b]$.

Bound that contribution.

Riemann integrability includes boundedness. Choose $B \geq 0$ with $|f(x)| \leq B$ on $[a, b]$. Restrictions of an integrable function are integrable, and additivity and the integral inequality give

$$\left| \int_a^b f - \int_a^d f \right| = \left| \int_d^b f \right| \leq \int_d^b |f| \leq B(b - d).$$

If $B = 0$, the difference is already zero. If $B > 0$, given $\varepsilon > 0$, choosing $0 < b - d < \varepsilon/B$ makes the difference less than ε.

Conclusion. The difference tends to zero, so the two definitions agree.

36.2 Exercise 36.2: The splitting point does not change the value

What we must show. Show that the definition of an integral on (a, b) is independent of the chosen interior point.

Compare two splitting points by a finite integral.

Suppose $a < \alpha < \beta < b$ and set $C = \int_\alpha^\beta f$. This is a finite ordinary integral by the local integrability hypothesis. For $a < c < \alpha$ and $\beta < d < b$,

$$\int_c^\beta f = \int_c^\alpha f + C, \qquad \int_\beta^d f = \int_\alpha^d f - C.$$

Pass to the endpoint limits. Adding or subtracting a finite number preserves the existence and sign of an infinite limit. Thus, if $A = \int_a^\alpha f$ and $B = \int_\alpha^b f$ exist, then

$$\int_a^\beta f = A + C, \qquad \int_\beta^b f = B - C.$$

Check that the permitted sum is unchanged.

If A, B are finite, their sum is plainly unchanged. If one is infinite and their sum is permitted, the finite changes do not alter either infinite sign. An excluded sum $+\infty + (-\infty)$ remains excluded. The same argument in reverse proves the converse, and interchanging α, β covers either order.

Conclusion. The existence and the value of the improper integral do not depend on the split.

36.3 Exercise 36.3: Power singularities at zero and at infinity

What we must show. Analyze $\int_0^1 x^{-p}\,dx$ and $\int_0^\infty x^{-p}\,dx$ for $p > 0$.

(a) Cut off the singular endpoint before integrating.

For $0 < c < 1$ and $p \neq 1$, the power rule gives

$$\int_c^1 x^{-p}\,dx = \frac{1 - c^{1-p}}{1-p}.$$

If $0 < p < 1$, the exponent $1 - p$ is positive, so $c^{1-p} \to 0$; the answer is $1/(1-p)$. If $p > 1$, then $c^{1-p} = 1/c^{p-1} \to +\infty$. Rewriting the last display as $(c^{1-p} - 1)/(p-1)$ shows that the integral tends to $+\infty$. For completeness, the boundary case $p = 1$ has $\int_c^1 dx/x = -\log c \to +\infty$.

(b) Treat both ends separately.

For $p \neq 1$,

$$\int_1^d x^{-p}\,dx = \frac{d^{1-p} - 1}{1-p}.$$

This tends to $1/(p-1)$ if $p > 1$, and to $+\infty$ if $0 < p < 1$. For $p = 1$ it equals $\log d \to +\infty$. Thus $p > 1$ fails at zero, $0 < p < 1$ fails at infinity, and $p = 1$ fails at both ends. All integrands are nonnegative, so there is no cancellation issue.

Conclusion. The integral on $(0, 1)$ converges exactly for $0 < p < 1$; the integral on $(0, \infty)$ is $+\infty$ for every $p > 0$.

36.4 Exercise 36.4: Three explicit improper integrals

(a) The logarithm near zero.

Integration by parts on $[c, 1]$, where $c > 0$, yields

$$\int_c^1 \log x\,dx = [x \log x - x]_c^1 = -1 - c\log c + c.$$

To justify $c \log c \to 0$, put $t = 1/c$. The elementary estimate $\log t = \int_1^t du/u \le \int_1^t du/\sqrt{u} = 2(\sqrt{t} - 1)$ for $t \ge 1$ gives $0 \le (\log t)/t \le 2/\sqrt{t} \to 0$. Consequently $\int_0^1 \log x\,dx = -1$.

(b) A logarithm divided by its argument.

For $d > 2$, substitution $u = \log x$, $du = dx/x$, gives

$$\int_2^d \frac{\log x}{x}\,dx = \frac{(\log d)^2 - (\log 2)^2}{2} \longrightarrow +\infty.$$

(c) The reciprocal quadratic.

Since $(\arctan x)' = 1/(1+x^2)$,

$$\int_0^d \frac{dx}{1+x^2} = \arctan d \longrightarrow \frac{\pi}{2}.$$

Conclusion. The values, in order, are -1, $+\infty$, and $\pi/2$.

36.5 Exercise 36.5: A nonnegative integral as a supremum

What we must show. For continuous $f \geq 0$ on (a, b), identify its improper integral with the supremum of all integrals over compact subintervals.

Establish the two one-sided limits.

Choose $\alpha \in (a, b)$. Define

$$A(c) = \int_c^\alpha f \quad (a < c < \alpha), \qquad B(d) = \int_\alpha^d f \quad (\alpha < d < b).$$

Because $f \geq 0$, $A(c)$ increases as c moves toward a, and $B(d)$ increases as d moves toward b. The monotone-limit property therefore gives limits $A, B \in [0, \infty]$, equal to their respective suprema. The improper integral exists and equals $A + B$; this sum can never have the forbidden form $+\infty + (-\infty)$.

Compare with all compact subintervals.

Let $M = \sup\{\int_c^d f : [c, d] \subset (a, b)\}$. Enlarging any $[c, d]$ so that it contains α can only increase its integral. An enlarged interval has integral $A(c') + B(d') \leq A + B$; thus $M \leq A + B$. Conversely, all intervals with $c < \alpha < d$ are allowed in the supremum, so $M \geq A(c) + B(d)$. If A, B are finite, choose c, d making these terms within $\varepsilon/2$ of their suprema, and obtain $M \geq A + B - \varepsilon$. If either supremum is infinite, these same integrals exceed every prescribed bound, so $M = +\infty$.

Conclusion. $\displaystyle\int_a^b f = \sup_{[c,d]\subset(a,b)} \int_c^d f$, with $+\infty$ allowed.

36.6 Exercise 36.6: Comparison tests

What we must show. Suppose $0 \leq f \leq g$ and both functions are continuous on (a, b). Prove the convergence and divergence comparisons.

Compare finite intervals first.

For every compact $[c, d] \subset (a, b)$, ordinary integral monotonicity gives $0 \leq \int_c^d f \leq \int_c^d g$. Exercise 36.5 allows us to take suprema, giving

$$0 \leq \int_a^b f \leq \int_a^b g \leq +\infty.$$

(a) Finite upper bound.

If $\int_a^b g$ is finite, the middle quantity is bounded above by that finite number. The nonnegative integral of f therefore converges.

(b) Infinite lower bound.

If $\int_a^b f = +\infty$, no finite number can bound the integral of g from above. Therefore $\int_a^b g = +\infty$.

Conclusion. A smaller nonnegative function inherits convergence; a larger one inherits divergence to $+\infty$.

36.7 Exercise 36.7: The Gaussian integral

(a) Prove finiteness before attempting evaluation.

The integrand is even. On $[0,1]$ it is at most 1; on $[1,\infty)$, $x^2 \geq x$, so $e^{-x^2} \leq e^{-x}$. The comparison test gives

$$0 < I := \int_{-\infty}^{\infty} e^{-x^2}\,dx \leq 2\left(1 + \int_1^{\infty} e^{-x}\,dx\right) = 2(1+e^{-1}) < \infty.$$

(b) Explain the double-integral calculation.

This part uses polar coordinates for a double integral, as requested by the exercise. The following bounded-region argument avoids assuming in advance that improper iterated integrals may be interchanged. Set $I_R = \int_{-R}^{R} e^{-x^2}\,dx$. On the square $Q_R = [-R,R]^2$, integration of the continuous product gives

$$\iint_{Q_R} e^{-x^2-y^2}\,dx\,dy = I_R^2.$$

Let D_R be the disk of radius R. The inclusions $D_R \subset Q_R \subset D_{\sqrt{2}R}$ and positivity give a lower and an upper bound for I_R^2. In polar coordinates,

$$\iint_{D_R} e^{-x^2-y^2}\,dx\,dy = \int_0^{2\pi}\int_0^R e^{-r^2} r\,dr\,d\theta = \pi(1-e^{-R^2}).$$

Here the extra factor r is the polar area factor, and $\int_0^R e^{-r^2} r\,dr = (1-e^{-R^2})/2$. Hence

$$\pi(1-e^{-R^2}) \leq I_R^2 \leq \pi(1-e^{-2R^2}).$$

Both outer expressions tend to π. Since part (a) shows $I_R \to I$, we obtain $I^2 = \pi$; and $I > 0$ selects the positive root.

Conclusion. $\displaystyle\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$.

36.8 Exercise 36.8: Absolute convergence implies convergence

What we must show. If f is continuous on (a,b) and $\int_a^b |f| < \infty$, prove that $\int_a^b f$ exists and is finite.

Plan. Write f as the difference of two nonnegative functions whose integrals are bounded above by integrals already known to converge.

Use positive and negative parts.

Define

$$f_+(x) = \max\{f(x), 0\}, \qquad f_-(x) = \max\{-f(x), 0\}.$$

These are continuous because, for example, $f_+ = (|f| + f)/2$. Pointwise, $0 \le f_+, f_- \le |f|$ and $f = f_+ - f_-$. Exercise 36.6 proves that both nonnegative improper integrals are finite.

Pass to limits without subtracting infinities.

On each compact subinterval, ordinary linearity gives $\int_c^d f = \int_c^d f_+ - \int_c^d f_-$. Fix an interior split, and take the left and right endpoint limits separately. Each term has a finite limit, so the limit law for subtraction applies.

Conclusion. $\int_a^b f$ converges and $\left|\int_a^b f\right| \le \int_a^b |f|$.

36.9 Exercise 36.9: Integrals for the normal distribution

Write $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ and $F(t) = \int_{-\infty}^t \varphi(x)\,dx$. Substitution $x = \sqrt{2}u$ in Exercise 36.7 gives $\int_{\mathbb{R}} \varphi = 1$.

(a) Replace the Stieltjes integrator by its density.

The Fundamental Theorem of Calculus gives $F' = \varphi$, a continuous function. For continuous f, Theorem 35.13 on each finite $[a, b]$ therefore yields

$$\int_a^b f\,dF = \int_a^b f(x)F'(x)\,dx = \frac{1}{\sqrt{2\pi}} \int_a^b f(x)e^{-x^2/2}\,dx.$$

Taking the two endpoint limits separately proves the same identity on $\mathbb{R}$ whenever the integral specified in the question exists; positive constant multiplication preserves finite and infinite limits.

(b) The second moment.

Since $\varphi'(x) = -x\varphi(x)$, integration by parts on $[a, b]$ gives

$$\int_a^b x^2\varphi(x)\,dx = [-x\varphi(x)]_a^b + \int_a^b \varphi(x)\,dx.$$

The boundary term tends to zero at either infinity. For instance, $e^t \ge t^2/2$ for $t \ge 0$ implies $xe^{-x^2/2} \le 8/x^3$ for $x > 0$, and negative x are handled by absolute values. Thus $\int_{\mathbb{R}} x^2\,dF = 1$.

(c) The exponential of the square.

The integrand here is e^{x^2}, not e^x. Its density product is

$$e^{x^2}\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{x^2/2} \ge \frac{1}{\sqrt{2\pi}}.$$

The integral of this positive constant over an infinite half-line is infinite, so comparison proves $\int_{\mathbb{R}} e^{x^2}\,dF = +\infty$.

(d) The absolute first moment.

Evenness and the antiderivative of $xe^{-x^2/2}$ give

$$\int_{\mathbb{R}} |x|\,dF = \frac{2}{\sqrt{2\pi}} \int_0^\infty xe^{-x^2/2}\,dx = \frac{2}{\sqrt{2\pi}}[-e^{-x^2/2}]_0^\infty = \sqrt{\frac{2}{\pi}}.$$

(e) The first moment and a necessary convergence check.

Part (d) shows absolute convergence. Each half-integral is therefore finite. Because $x\varphi(x)$ is odd, substitution $u = -x$ in the negative half-integral makes it the negative of the positive half. Their sum is zero. Oddness alone would not suffice without the existence of both halves.

Conclusion. The values in (b)–(e) are 1, $+\infty$, $\sqrt{2/\pi}$, and 0.

36.10 Exercise 36.10: Linearity on the whole real line

What we must show. For F-integrable f, g on $\mathbb{R}$, prove integrability and the sum rule for $f + g$.

Apply finite-interval linearity.

For every finite $a < 0 < b$, Theorem 35.8 gives

$$\int_a^0 (f+g)\,dF = \int_a^0 f\,dF + \int_a^0 g\,dF, \qquad \int_0^b (f+g)\,dF = \int_0^b f\,dF + \int_0^b g\,dF.$$

Since f, g are F-integrable on $\mathbb{R}$, all four half-line limits are finite. The limit law for addition proves that the two limits for $f + g$ exist and are finite. Add them and regroup finite real numbers.

Conclusion. $f + g$ is F-integrable and $\int_{\mathbb{R}}(f+g)\,dF = \int_{\mathbb{R}} f\,dF + \int_{\mathbb{R}} g\,dF$. The same argument proves scalar multiplication, including multiplication by -1.

36.11 Exercise 36.11: Monotonicity on the whole real line

What we must show. If f, g are F-integrable and $f \le g$, prove the integral inequality.

Preserve the inequality through finite limits.

Theorem 35.9 gives $\int_a^0 f\,dF \le \int_a^0 g\,dF$ for each $a < 0$, and $\int_0^b f\,dF \le \int_0^b g\,dF$ for each $b > 0$. Both sides have finite limits by hypothesis. If real-valued quantities satisfy an inequality before taking a limit, the limits satisfy the same weak inequality. Pass to $a \to -\infty$ and $b \to +\infty$ separately, and add.

Conclusion. $\displaystyle\int_{\mathbb{R}} f\,dF \le \int_{\mathbb{R}} g\,dF.$

36.12 Exercise 36.12: Comparison for improper Stieltjes integrals

What we must show. State and prove the analogue of Exercise 36.6. Assume F is increasing and f, g are F-integrable on every finite interval, with $0 \le f \le g$. Continuity of f, g is one sufficient condition for local integrability.

Nonnegativity supplies the endpoint limits.

For $b > 0$, $\int_0^b f\,dF$ is increasing in b, since its increment over an added interval is nonnegative. Likewise, $\int_a^0 f\,dF$ increases as $a \to -\infty$. Thus both limits exist in $[0, \infty]$. The same holds for g.

Compare before and after the limits.

Finite-interval monotonicity gives

$$0 \le \int_a^0 f\,dF \le \int_a^0 g\,dF, \qquad 0 \le \int_0^b f\,dF \le \int_0^b g\,dF.$$

Taking limits and adding nonnegative extended values proves $0 \le \int_{\mathbb{R}} f\,dF \le \int_{\mathbb{R}} g\,dF$.

Conclusion. If $\int_{\mathbb{R}} g\,dF < \infty$, then $\int_{\mathbb{R}} f\,dF < \infty$. If $\int_{\mathbb{R}} f\,dF = +\infty$, then $\int_{\mathbb{R}} g\,dF = +\infty$.

36.13 Exercise 36.13: Absolute convergence for Stieltjes integrals

What we must show. If f is continuous on $\mathbb{R}$ and $\int_{\mathbb{R}} |f|\,dF < \infty$, prove f is F-integrable on $\mathbb{R}$.

Obtain finite nonnegative integrals.

Continuity makes f, $|f|$, and $h = f + |f|$ locally F-integrable. Pointwise, $0 \le h \le 2|f|$. Exercise 36.12 shows that h has a finite integral, since $2|f|$ does.

Subtract a function with a known finite integral.

Scalar multiplication shows $-|f|$ is F-integrable. Exercise 36.10 now applies to $f = h + (-|f|)$, so f is F-integrable. Applying monotonicity to $-|f| \le f \le |f|$ also gives $|\int_{\mathbb{R}} f\,dF| \le \int_{\mathbb{R}} |f|\,dF$.

Conclusion. Absolute F-integrability implies F-integrability. The argument still works if continuity is replaced by local F-integrability of f and $|f|$.

36.14 Exercise 36.14: A countable collection of point masses

What we must show. Let the distinct points be $u_j \in \mathbb{R}$, let $c_j \ge 0$, and suppose $C = \sum_{j=1}^{\infty} c_j < \infty$. Analyze $F(t) = \sum_j c_j J_{u_j}(t)$, where $J_u(t) = 0$ for $t < u$ and $J_u(t) = 1$ for $t \ge u$ on $\mathbb{R}$.

(a) Existence and monotonicity.

For each t, $0 \le c_j J_{u_j}(t) \le c_j$. The series therefore converges, and its tail is at most $\sum_{j>N} c_j$, independently of t. Thus convergence is uniform. Each summand is increasing; for $s < t$, termwise comparison gives $F(s) \le F(t)$.

(b) Finite intervals and their endpoint convention.

Let $|f| \le B$ on $\mathbb{R}$. Theorem 35.12(b), or its finite-mass tail estimate, makes every bounded f F-integrable on $[a, b]$. For this right-continuous F, the formula is

$$\int_a^b f\,dF = \sum_{j:\, a<u_j\le b} c_j f(u_j).$$

Indeed a single global step contributes its mass exactly when $a < u_j \le b$; a jump at a has already occurred before the interval's increment begins. Summing the steps is justified because the absolute value of the contribution from all indices $j > N$ is at most $B \sum_{j>N} c_j$. This tends to zero.

Pass to the whole line, controlling the tails.

The numerical series $\sum_j c_j f(u_j)$ is absolutely convergent, because $\sum_j |c_j f(u_j)| \le BC$. To see directly that the finite-interval sums approach it, choose N with $B \sum_{j>N} c_j < \varepsilon$ (the case $B = 0$ is immediate), then choose $a < \min_{j\le N} u_j$ and $b > \max_{j\le N} u_j$. All first N terms are included, and the omitted terms have total absolute value below ε. The same argument on the two halves treats $u_j = 0$ in the negative half only. Hence

$$\int_{-\infty}^{\infty} f\,dF = \sum_{j=1}^{\infty} c_j f(u_j).$$

(c) The total mass condition.

The same finite-head/small-tail argument gives $F(-\infty) = 0$ and $F(\infty) = C$. Consequently F is a distribution function exactly when $\sum_j c_j = 1$. It is also right-continuous: each finite partial sum is right-continuous, and uniform convergence preserves this one-sided continuity.

(d) Dense jumps and a necessary qualification.

For $s < t$,

$$F(t) - F(s) = \sum_{j:\, s<u_j\le t} c_j.$$

If the u_j enumerate $\mathbb{Q}$ and *every* $c_j > 0$, there is a rational u_j strictly between s and t, so this difference is positive. Thus F is strictly increasing. More generally the exact condition is that $\{u_j : c_j > 0\}$ is dense in $\mathbb{R}$.

Conclusion. Parts (a)–(c) hold for nonnegative coefficients as stated. Part (d), read with merely nonnegative coefficients, needs the additional density condition: all $c_j = 0$ would give $F = 0$. This is the positive-coefficient situation of Exercise 35.12.

36.15 Exercise 36.15: Uniform convergence on spaces of infinite and finite mass

(a) Small height can be offset by large width.

Define $f_n(x) = 1/n$ on $[0, n]$, and $f_n(x) = 0$ elsewhere. Each function is piecewise constant and has a finite improper Riemann integral. Its rectangle has height $1/n$ and width n, so

$$\int_{\mathbb{R}} f_n(x)\,dx = 1, \qquad \sup_{x\in\mathbb{R}} |f_n(x) - 0| = \frac{1}{n} \longrightarrow 0.$$

Thus uniform convergence alone does not preserve an integral over a domain of infinite ordinary length.

(b) Local integrability of the uniform limit.

Now suppose F is a distribution function, all f_n are F-integrable on $\mathbb{R}$, and $f_n \to f$ uniformly on $\mathbb{R}$. On each finite $[a, b]$, the uniform-limit theorem for Stieltjes integrals (Exercise 35.6) makes f F-integrable there. For clarity, that theorem follows from

$$U_F(f, P) - L_F(f, P) \le U_F(f_n, P) - L_F(f_n, P) + 2\|f - f_n\|_\infty[F(b) - F(a)] :$$

first choose n to make the second term small, then choose P using integrability of f_n.

Prove global integrability without assuming global boundedness.

Choose one index N with $\|f - f_N\|_\infty \le 1$. The difference $h = f - f_N$ is bounded and locally F-integrable. Since F has finite total increment, Theorem 36.4 makes h F-integrable on $\mathbb{R}$. Exercise 36.10 now shows that $f = f_N + h$ is F-integrable. This argument does not require f_N, or f, to be bounded on all of $\mathbb{R}$.

Estimate the difference of the integrals.

For any sufficiently large n, put $\eta_n = \|f - f_n\|_\infty$. By monotonicity and the total-mass identity $\int_{\mathbb{R}} 1\,dF = F(\infty) - F(-\infty) = 1$,

$$-\eta_n \le \int_{\mathbb{R}} (f - f_n)\,dF \le \eta_n.$$

Linearity therefore gives

$$\left| \int_{\mathbb{R}} f \, dF - \int_{\mathbb{R}} f_n \, dF \right| \leq \eta_n \longrightarrow 0.$$

Conclusion. For a distribution function, uniform convergence does preserve the integral. The decisive bound is the finite total mass 1, rather than the geometric length of the real line.